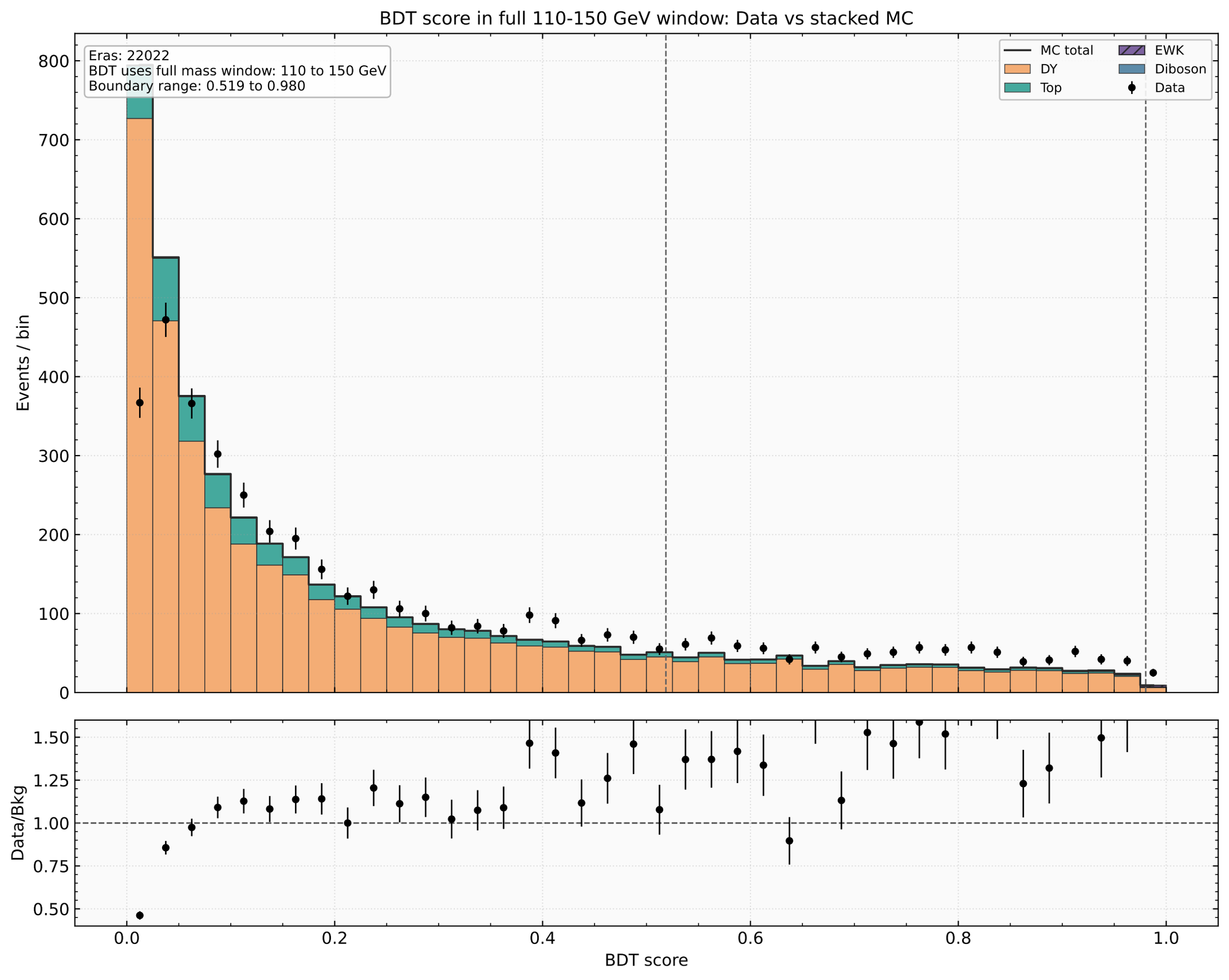
Run-3 BDT Sideband Validation Report

Outputs directory: /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Eras used: 22022
Extra cut: (none)
Blinding window: 115 < m_mumu < 135 GeV
Sidebands: (110, 115) U (135, 150) GeV
Optimized category edges: 0.0000, 0.5187, 0.7693, 0.8902, 0.9502, 0.9803, 1.0000

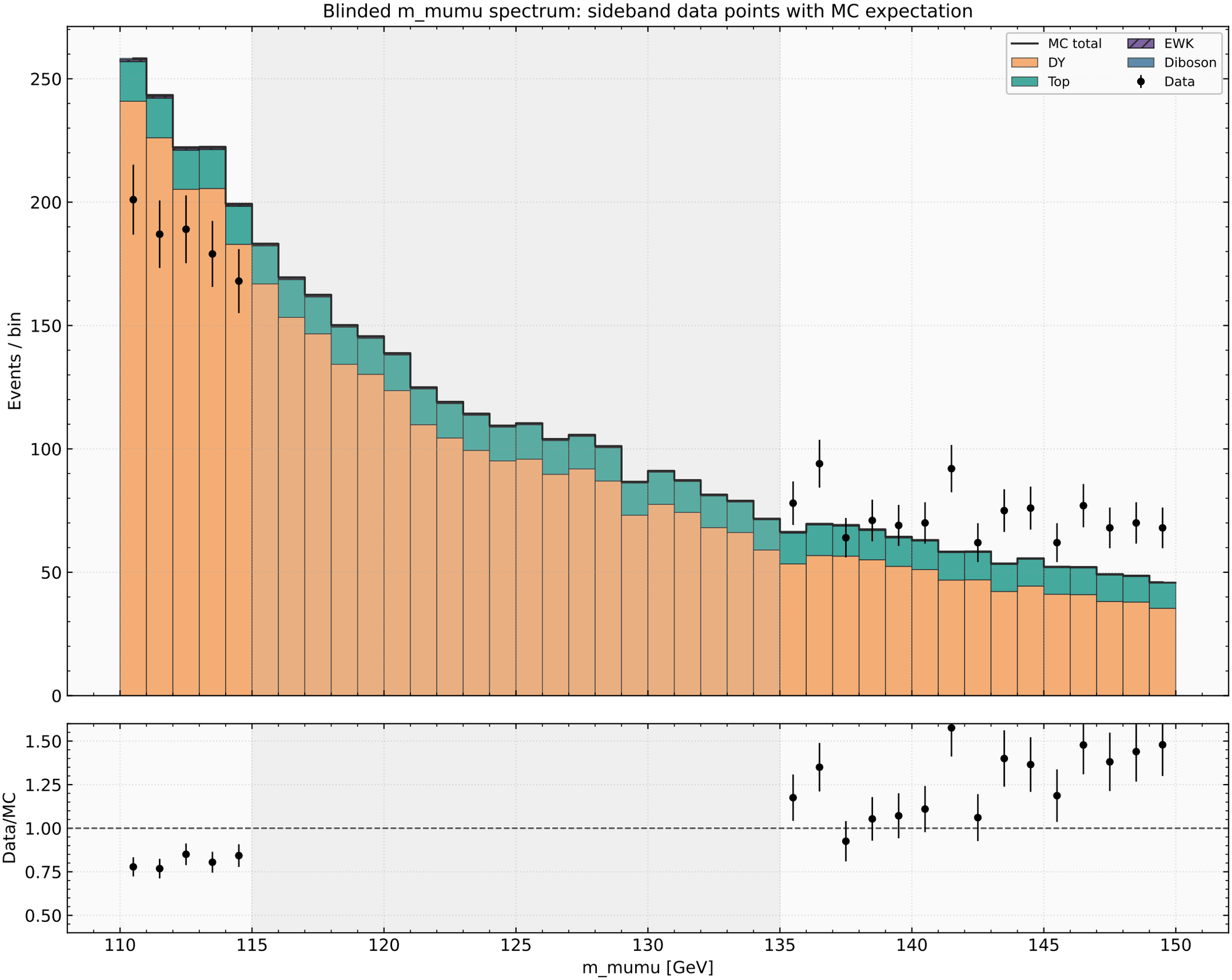

BDT score in full 110-150 GeV window: Data vs stacked MC
Eras: 22022
BDT uses full mass window: 110 to 150 GeV
Boundary range: 0.519 to 0.980
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data/Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

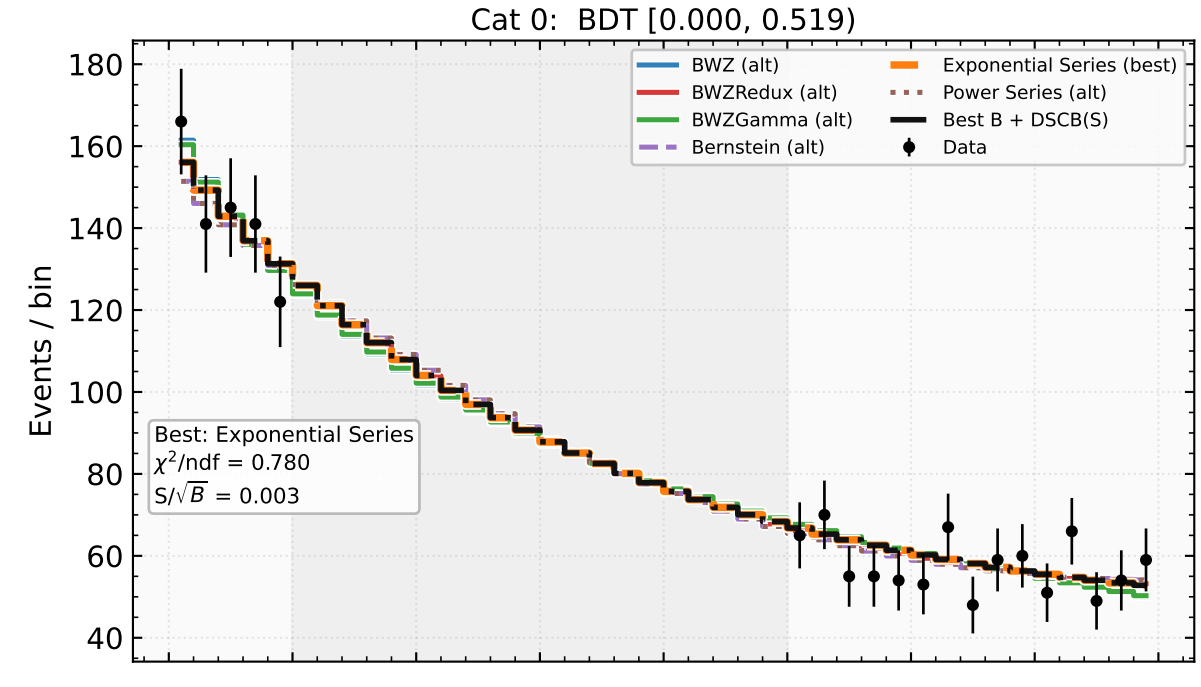
BDT bin	Total signal	Bkg. ± DeltaB	Data	S/(S+B) (%)	S/sqrt(B)
[0.000, 0.025)	0.003	794.83 ± 12.01	367	0.00	0.000
[0.025, 0.050)	0.007	551.25 ± 9.63	472	0.00	0.000
[0.050, 0.075)	0.008	375.63 ± 7.83	366	0.00	0.000
[0.075, 0.100)	0.008	276.86 ± 6.59	302	0.00	0.000
[0.100, 0.125)	0.008	221.74 ± 5.95	250	0.00	0.001
[0.125, 0.150)	0.008	188.59 ± 5.52	204	0.00	0.001
[0.150, 0.175)	0.008	171.46 ± 5.27	195	0.00	0.001
[0.175, 0.200)	0.008	136.74 ± 4.66	156	0.01	0.001
[0.200, 0.225)	0.008	121.99 ± 4.37	122	0.01	0.001
[0.225, 0.250)	0.009	107.93 ± 4.17	130	0.01	0.001
[0.250, 0.275)	0.008	95.28 ± 3.88	106	0.01	0.001
[0.275, 0.300)	0.009	86.95 ± 3.76	100	0.01	0.001
[0.300, 0.325)	0.009	80.16 ± 3.63	82	0.01	0.001
[0.325, 0.350)	0.009	78.20 ± 3.50	84	0.01	0.001
[0.350, 0.375)	0.010	71.61 ± 3.36	78	0.01	0.001
[0.375, 0.400)	0.010	66.90 ± 3.23	98	0.01	0.001
[0.400, 0.425)	0.010	64.61 ± 3.24	91	0.02	0.001
[0.425, 0.450)	0.011	59.11 ± 3.01	66	0.02	0.001
[0.450, 0.475)	0.011	57.91 ± 3.07	73	0.02	0.001
[0.475, 0.500)	0.012	47.94 ± 2.78	70	0.02	0.002

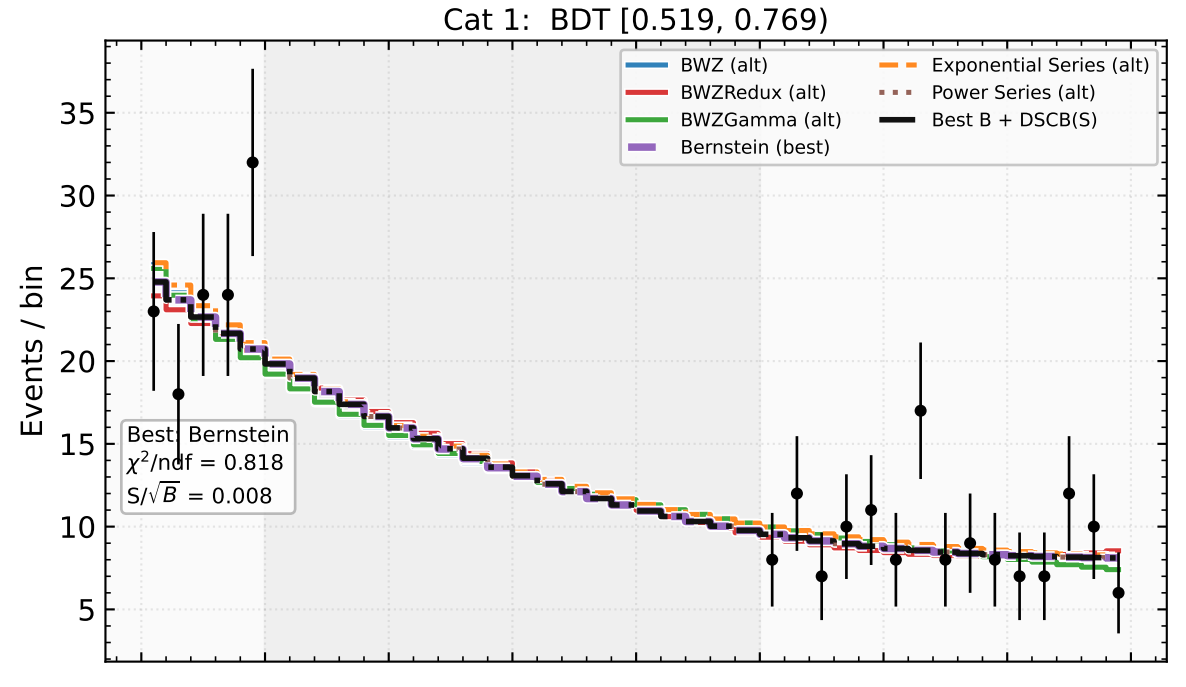
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

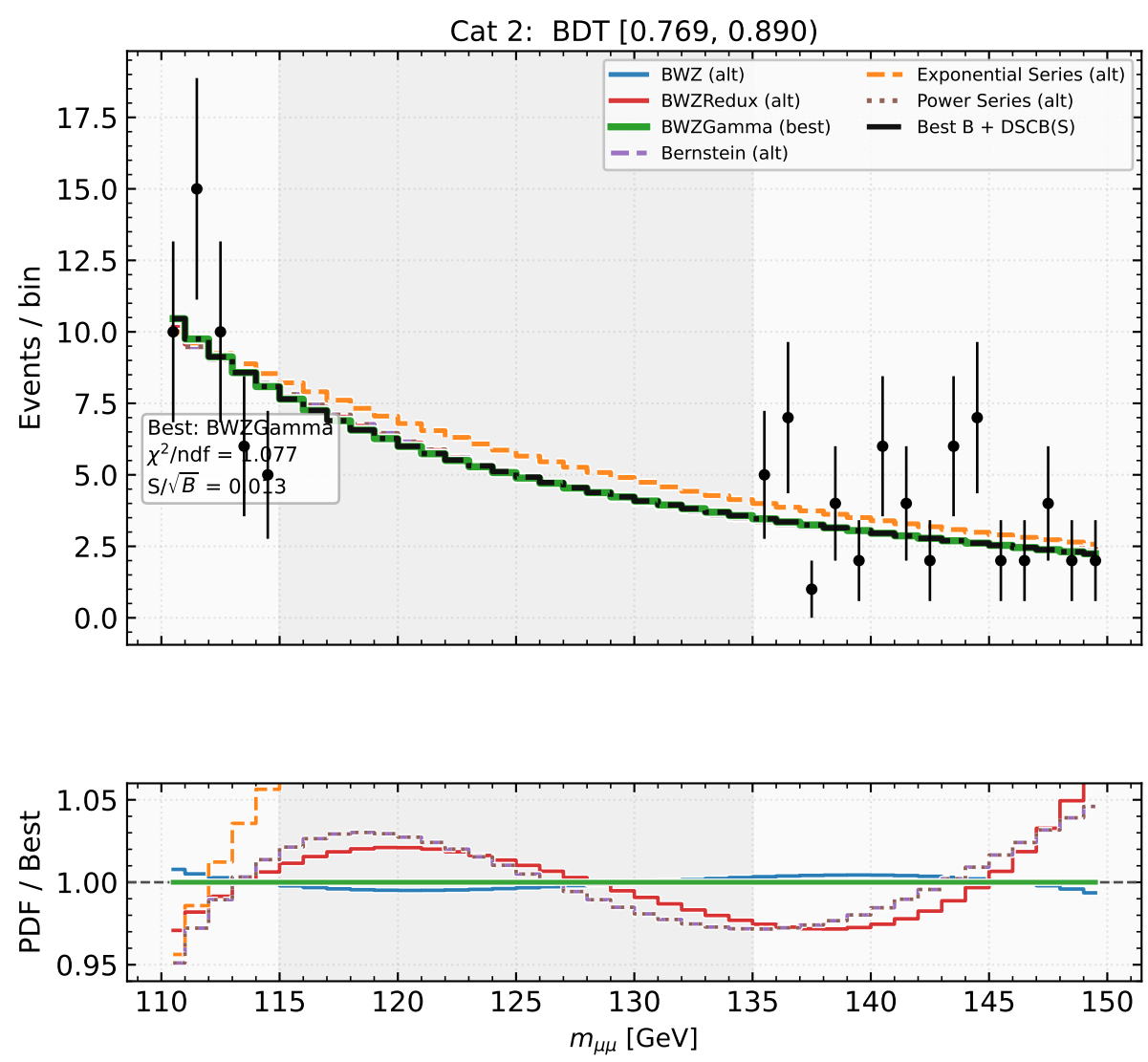
BDT bin	Total signal	Bkg. ± DeltaB	Data	S/(S+B) (%)	S/sqrt(B)
[0.500, 0.525)	0.013	51.04 ± 2.82	55	0.02	0.002
[0.525, 0.550)	0.013	44.52 ± 2.66	61	0.03	0.002
[0.550, 0.575)	0.014	50.33 ± 2.81	69	0.03	0.002
[0.575, 0.600)	0.015	41.62 ± 2.55	59	0.04	0.002
[0.600, 0.625)	0.016	41.88 ± 2.55	56	0.04	0.002
[0.625, 0.650)	0.016	46.86 ± 2.77	42	0.03	0.002
[0.650, 0.675)	0.018	33.83 ± 2.27	57	0.05	0.003
[0.675, 0.700)	0.020	39.76 ± 2.50	45	0.05	0.003
[0.700, 0.725)	0.021	32.08 ± 2.17	49	0.07	0.004
[0.725, 0.750)	0.023	34.87 ± 2.29	51	0.07	0.004
[0.750, 0.775)	0.025	35.90 ± 2.40	57	0.07	0.004
[0.775, 0.800)	0.028	35.55 ± 2.32	54	0.08	0.005
[0.800, 0.825)	0.032	31.57 ± 2.17	57	0.10	0.006
[0.825, 0.850)	0.036	29.45 ± 2.06	51	0.12	0.007
[0.850, 0.875)	0.043	31.72 ± 2.22	39	0.13	0.008
[0.875, 0.900)	0.051	31.06 ± 2.17	41	0.16	0.009
[0.900, 0.925)	0.065	27.55 ± 2.01	52	0.23	0.012
[0.925, 0.950)	0.087	28.07 ± 2.05	42	0.31	0.016
[0.950, 0.975)	0.133	23.83 ± 1.82	40	0.56	0.027
[0.975, 1.000)	0.211	9.20 ± 1.01	25	2.24	0.070

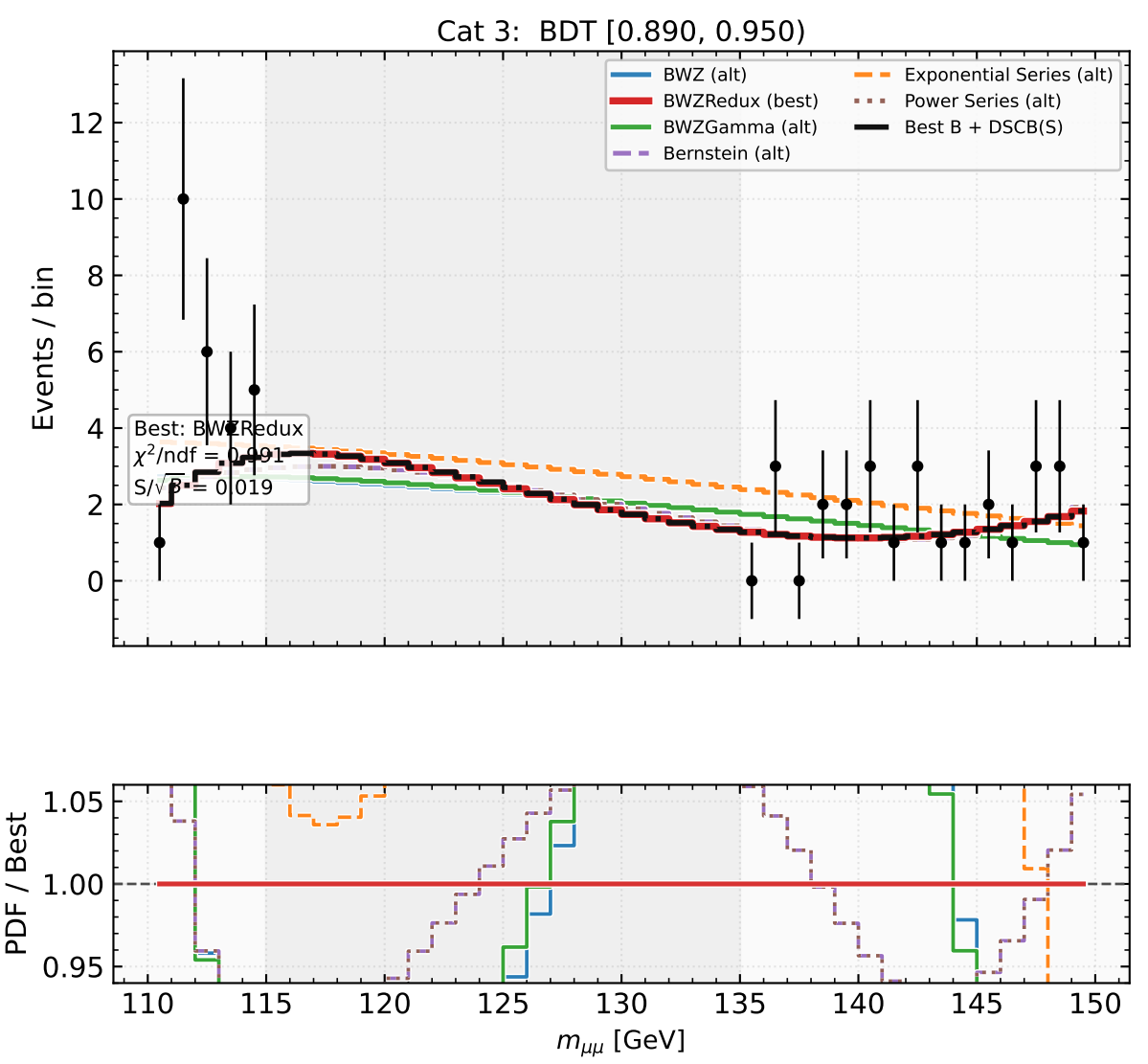

Blinded m_mumu spectrum: sideband data points with MC expectation
MC total
DY
Top
EWK
Diboson
Data
Events / bin
0
50
100
150
200
250
Data/MC
0.50
0.75
1.00
1.25
1.50
110
115
120
125
130
135
140
145
150
m_mumu [GeV]

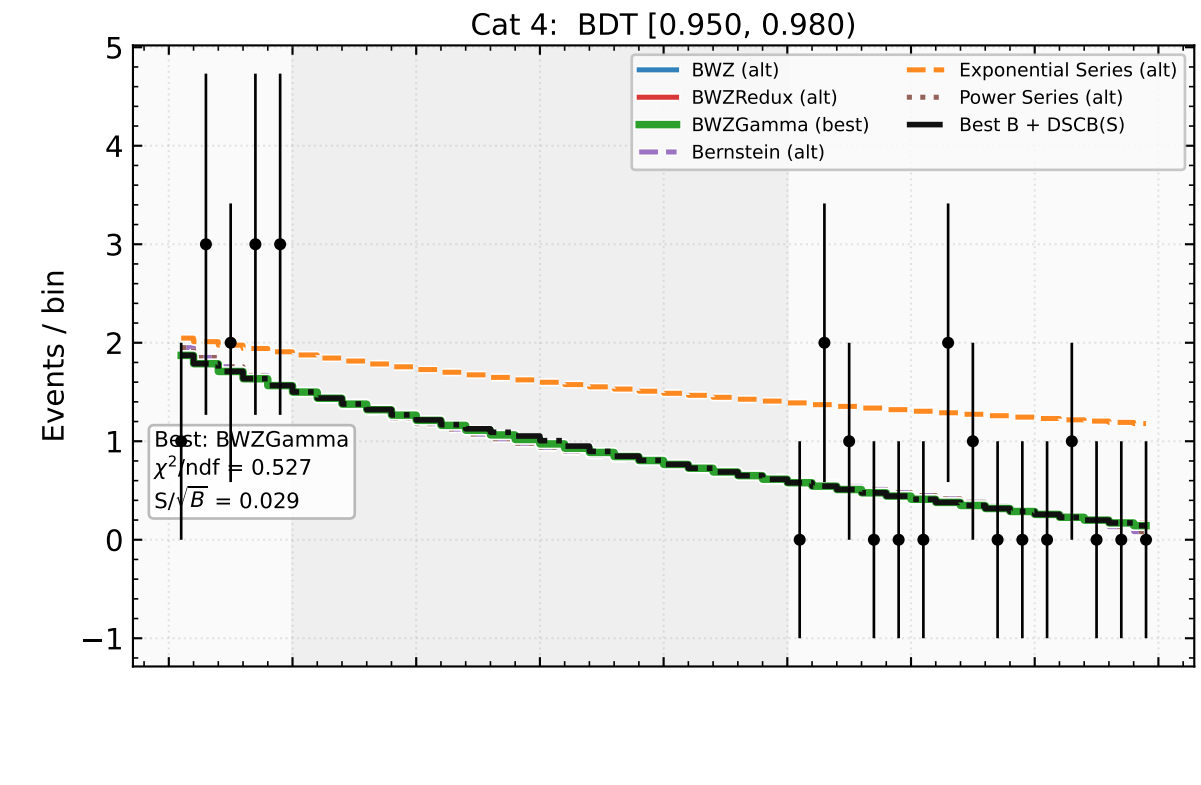
Per-category dimuon mass: background model comparison and ratio-to-best

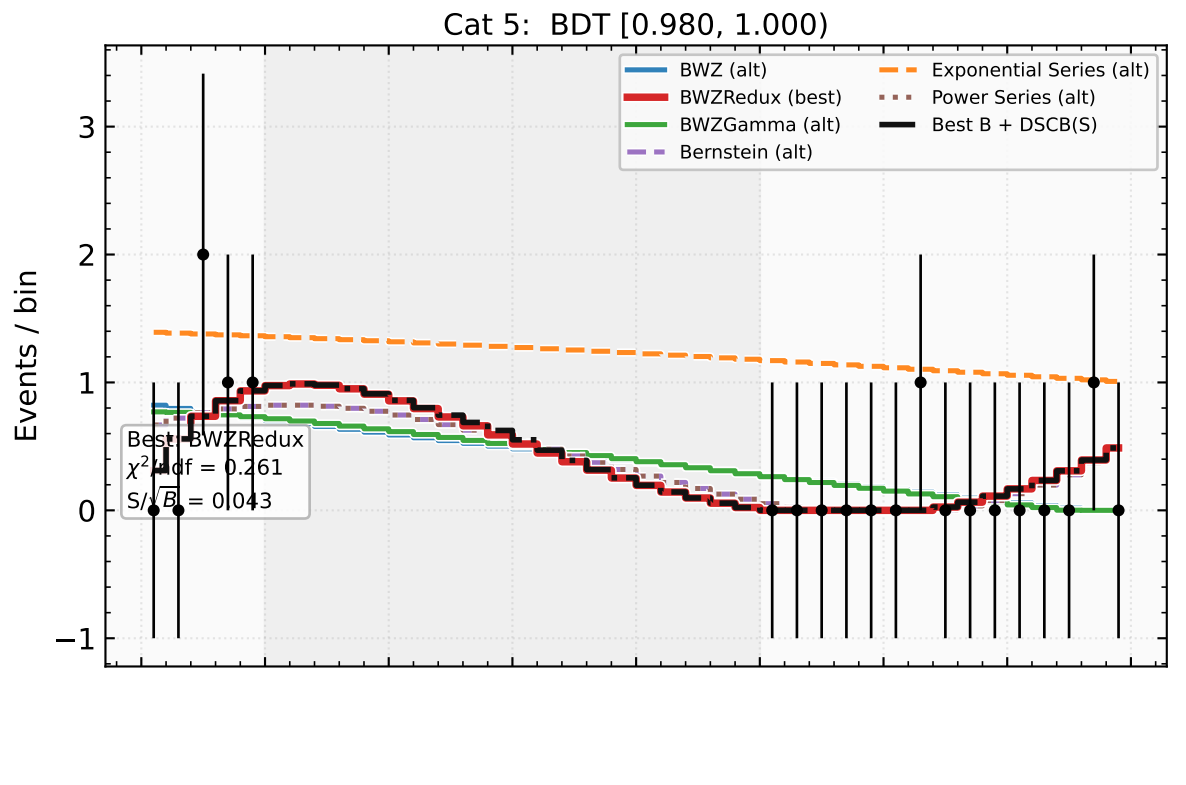












Per-category full-range fit comparison (chi2/ndf, ndf)

Category	BDT range	Best model	Best chi2/ndf	S/sqrt(B)	BWZ	BWZRedux	BWZGamma	Bernstein	Exponential Series	Power Series
0	[0.000, 0.519)	Exponential Series	0.780	0.003	0.800 (37)	0.782 (36)	0.791 (37)	0.804 (36)	0.780 (37)	0.804 (36)
1	[0.519, 0.769)	Bernstein	0.818	0.008	0.836 (37)	0.822 (36)	0.833 (37)	0.818 (36)	0.819 (37)	0.818 (36)
2	[0.769, 0.890)	BWZGamma	1.077	0.013	1.078 (37)	1.105 (36)	1.077 (37)	1.105 (36)	1.154 (37)	1.105 (36)
3	[0.890, 0.950)	BWZRedux	0.991	0.019	1.075 (37)	0.991 (36)	1.064 (37)	1.027 (36)	1.324 (37)	1.027 (36)
4	[0.950, 0.980)	BWZGamma	0.527	0.029	0.528 (37)	0.542 (36)	0.527 (37)	0.540 (36)	1.025 (37)	0.540 (36)
5	[0.980, 1.000)	BWZRedux	0.261	0.043	0.310 (37)	0.261 (36)	0.306 (37)	0.287 (36)	1.042 (37)	0.287 (36)

Optimized category summary

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	Expected signif. (S/sqrt(B))
0	[0.000, 0.519)	1580	1712.47 ± 16.76	0.923	3694.24 ± 24.75	0.003
1	[0.519, 0.769)	261	186.93 ± 5.41	1.396	405.43 ± 7.96	0.009
2	[0.769, 0.890)	102	71.18 ± 3.26	1.433	153.54 ± 4.80	0.014
3	[0.890, 0.950)	52	34.16 ± 2.30	1.522	70.54 ± 3.27	0.021
4	[0.950, 0.980)	19	12.55 ± 1.35	1.514	26.48 ± 1.91	0.034
5	[0.980, 1.000)	6	2.71 ± 0.49	2.213	6.14 ± 0.78	0.069

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.519)	3221.50	3.09	462.29	7.37	612.66	3694.24
1	[0.519, 0.769)	359.80	2.71	41.55	1.38	63.31	405.43
2	[0.769, 0.890)	136.08	2.36	14.56	0.54	23.08	153.54
3	[0.890, 0.950)	62.24	2.25	5.80	0.25	10.66	70.54
4	[0.950, 0.980)	22.30	2.00	2.10	0.09	3.67	26.48
5	[0.980, 1.000)	3.91	1.69	0.52	0.02	0.60	6.14